



hydrocarbons

Name:.....

Date:.....

Which is NOT a concern surrounding the use of biofuels, such as ethanol

- a) More people are needed to produce biofuels than petrol and diesel
- b) Crops that could be used to feed people are used for biofuels instead
- c) Fossil fuels maybe used in the production of biofuels
- d) Ethanol is not a hydrocarbon because it contains oxygen as well as hydrogen and carbon

1a

Correct Answer: A B C D

Explanation:

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Burning pure hydrocarbons produces...

- a) Carbon dioxide + water vapour
- b) Carbon dioxide + hydrogen
- c) Hydrogen + water vapour
- d) Water vapour only

1a

Correct Answer: A B C D

Explanation:

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What is NOT a product when hydrocarbons burn with not enough oxygen...

- a) Solid particles called particulates
- b) Soot (carbon) and unburnt fuel
- c) Carbon monoxide
- d) Hydrogen

1a

Correct Answer: A B C D

Explanation:

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Hydrocarbons can be cracked into smaller molecules. Which is not true?

- a) The amounts of fractions produced by fractional distillation is not the same as the demand for each of the fractions
- b) Cracking hydrocarbons involves heating to vaporise them
- c) The vapours are either passed over a hot catalyst or mixed with steam and heated to a very high temperature
- d) This is an example of a displacement reaction

1b

Correct Answer: A B C D

Explanation:

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Adding an Alkane to bromine water will turn it from...

- a) Orange to colourless
- b) Purple to colourless
- c) Colourless to purple
- d) No change

1b

Correct Answer: A B C D

Explanation:

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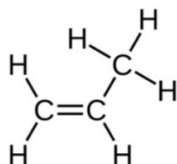
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What Alkene is this?



- a) Ethene
- b) Propene
- c) Ethane
- d) Propane

1b

Correct Answer: A B C D

Explanation:

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Alkenes are known as unsaturated because?

- a) Their double bond can open up allowing the two carbon atoms to bond with other atoms
- b) They can not react with water
- c) They do not have a double bond
- d) They can not be used to make polymers

1b

Correct Answer: A B C D

Explanation:

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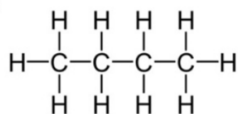
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What Alkane is this?



- a) Butene
- b) Propene
- c) Butane
- d) Propane

1b

Correct Answer: A B C D

Explanation:

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Alkenes are hydrocarbons that have?

- a) C=C double bond between two of the carbon atoms in their chain
- b) C-C single bond between two of the carbon atoms in their chain
- c) C=C double bond between all of the carbon atoms in their chain
- d) C-C single bond between all of the carbon atoms in their chain

1b

Correct Answer: A B C D

Explanation:

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Which is not true about cracking?

- a) Cracking is a thermal decomposition reaction
- b) Vapour is passed over a powered catalyst at a temperature of about 400-700°C
- c) Aluminium oxide is a catalyst used
- d) Short chains of hydrocarbon combine to form longer chains

1b

Correct Answer: A B C D

Explanation:

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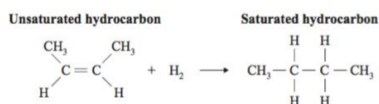
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Unsaturated oils contain double carbon carbon bonds. If an unsaturated hydrocarbon is heated with hydrogen gas and a nickel catalyst, the following reaction takes place.



What is this process called?

- A) cracking
- B) polymerisation
- C) hydrogenation
- D) combustion

Correct Answer: A B C D

Explanation:

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Which one of the following is the correct balanced equation for the polymerisation of ethene to form poly(ethene)?

- A)
$$n \begin{array}{c} \text{H} & \text{H} \\ | & | \\ \text{C} = & \text{C} \\ | & | \\ \text{H} & \text{H} \end{array} \longrightarrow \left[\begin{array}{c} \text{H} & \text{H} \\ | & | \\ -\text{C} = & \text{C}- \\ | & | \\ \text{H} & \text{H} \end{array} \right]_n$$
- B)
$$n \begin{array}{c} \text{H} & \text{H} \\ | & | \\ \text{C} = & \text{C} \\ | & | \\ \text{H} & \text{H} \end{array} \longrightarrow \left[\begin{array}{c} \text{H} & \text{H} \\ | & | \\ -\text{C} - & \text{C}- \\ | & | \\ \text{H} & \text{H} \end{array} \right]_n$$
- C)
$$\begin{array}{c} \text{H} & \text{H} \\ | & | \\ \text{C} = & \text{C} \\ | & | \\ \text{H} & \text{H} \end{array} \longrightarrow \left[\begin{array}{c} \text{H} & \text{H} \\ | & | \\ -\text{C} - & \text{C}- \\ | & | \\ \text{H} & \text{H} \end{array} \right]_n$$
- D)
$$\begin{array}{c} \text{H} & \text{H} \\ | & | \\ \text{C} = & \text{C} \\ | & | \\ \text{H} & \text{H} & n \end{array} \longrightarrow \left[\begin{array}{c} \text{H} & \text{H} \\ | & | \\ -\text{C} - & \text{C}- \\ | & | \\ \text{H} & \text{H} \end{array} \right]_n$$

Correct Answer: A B C D

Explanation:

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In which equation do the products include two different alkenes?

- A) $\text{C}_{15}\text{H}_{32} \rightarrow 2\text{C}_2\text{H}_4 + \text{C}_3\text{H}_6 + \text{C}_8\text{H}_{18}$
- B) $\text{C}_{15}\text{H}_{32} \rightarrow \text{C}_2\text{H}_2 + \text{C}_5\text{H}_{10} + \text{C}_8\text{H}_{18} + \text{H}_2$
- C) $\text{C}_{15}\text{H}_{32} \rightarrow \text{C}_8\text{H}_{18} + \text{C}_7\text{H}_{14}$
- D) $\text{C}_{15}\text{H}_{32} \rightarrow \text{C}_7\text{H}_{16} + \text{C}_8\text{H}_{16}$

Correct Answer: A B C D

Explanation:

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Which diagram represents and unsaturated hydrocarbon?

- a)
$$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$$
- b)
$$\begin{array}{c} \text{H} & \text{H} \\ | & | \\ \text{H}-\text{C}-\text{C}-\text{H} \\ | & | \\ \text{H} & \text{H} \end{array}$$
- c)
$$\left[\begin{array}{c} \text{H} & \text{H} \\ | & | \\ -\text{C}-\text{C}- \\ | & | \\ \text{H} & \text{H} \end{array} \right]_n$$
- d)
$$\begin{array}{c} \text{H} & \text{H} & & \text{H} \\ | & | & & / \\ \text{H}-\text{C}-\text{C}=\text{C} & & \backslash \\ | & & \text{H} \\ \text{H} & & \end{array}$$

Correct Answer: A B C D

Explanation:

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